A combined solid is said to be a solid formed by combining two or more solids.

The concept of combined solids is useful in the fields like doll making, building construction, carpentry, etc.

To calculate the surface area of the combined solid, we should only calculate the areas that are visible to our eyes. For example, if a cone is surmounted by a hemisphere, we need to just find out the C.S.A. of the hemisphere and C.S.A. of the cone separately and add them together. Note that we are leaving the base area of both the cone and the hemisphere since both the bases are attached together and are not visible.

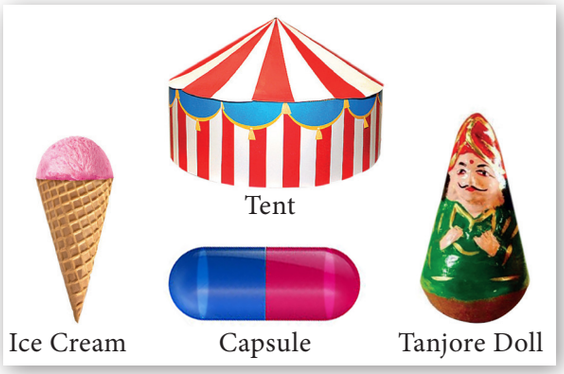


Fig. 7.37

But, the volume of the solid formed by joining two basic solids will be the sum of the volumes of the individual solids.

Example 7.24 A toy is in the shape of a cylinder surmounted by a hemisphere. The height of the toy is 25 cm. Find the total surface area of the toy if its common diameter is 12 cm.

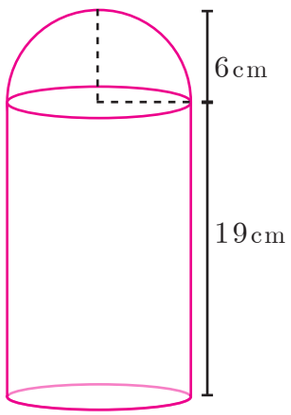


Fig. 7.38

Solution Let r and h be the radius and height of the cylinder respectively.

Given that, diameter $d = 12$ cm, radius $r = 6$ cm

Total height of the toy is 25 cm

Therefore, height of the cylindrical portion $= 25 - 6 = 19$ cm

T.S.A. of the toy = C.S.A. of the cylinder + C.S.A. of the hemisphere + Base Area of the cylinder

$$= 2\pi rh + 2\pi r^2 + \pi r^2$$

$$= \pi r(2h + 3r) \text{ sq. units}$$

$$= \frac{22}{7} \times 6 \times (38 + 18)$$

$$= \frac{22}{7} \times 6 \times 56 = 1056$$

Therefore, T.S.A. of the toy is 1056 cm^2

Example 7.25 A jewel box (Fig. 7.39) is in the shape of a cuboid of dimensions 30 cm $\times$ 15 cm $\times$ 10 cm surmounted by a half part of a cylinder as shown in the figure. Find the volume of the box.



Fig. 7.39

Solution Let l, b and h_1 be the length, breadth and height of the cuboid. Also let us take r and h_2 be the radius and height of the cylinder.

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